ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Foundations of the Landauer-Büttiker Transport Formalism

I. The Mesoscopic Paradigm Shift and The Origin of R_0

II. Derivation of the Two-Terminal Landauer Formula

III. Generalization to Multi-Terminal Networks and Topological Applications

I. The Mesoscopic Paradigm Shift and The Origin of R_0

The Limits of Ohm's Law: Length Scales in Mesoscopic Physics

Mean Free Path: Average distance an electron travels before losing its momentum to elastic scattering (impurities, defects)

$L \gg L_e$ and L_ϕ **Diffusive type: scattering dominated transport by impurities, phonons, or defects**

$L \ll L_e$ and L_ϕ **Ballistic type: no scattering transport and resistance independent of length**

Phase Coherence Length: Distance over which an electron maintains its phase memory before undergoing inelastic scattering (phonons, electron-electron interactions)

To understand why a new theory is needed, the physical dimensions of the channel (L) must be analyzed relative to the intrinsic length scales of the electron. In traditional macroscopic semiconductors, electrons behave like classical particles bumper-car-riding through a lattice of impurities. However, when the channel length L becomes smaller than the momentum relaxation length (L_e) and the phase coherence length (L_ϕ), electrons pass through the conductor without losing their wave-like quantum state. This is the ballistic transport regime

I. The Mesoscopic Paradigm Shift and The Origin of R_0

Thought experiment: A perfectly ordered 1D carbon nanotube or quantum wire with zero defects, zero phonons, and infinite crystal purity

Classical expectation based on Drude theory: Resistivity $R = {}^{\rho L}/_{A} = 0$

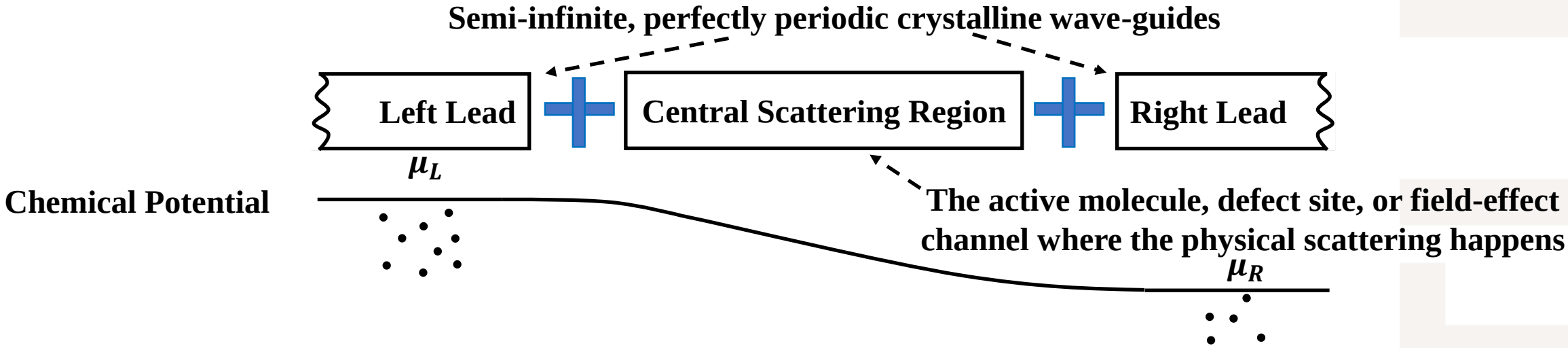
In reality: The experimental measured resistance does not approach zero. Instead, it hits a hard fundamental lower bound known as the contact resistance: $R = {}^{h}/_{2e^2} \approx 12.9\ k\Omega$

Consider a flawless 1D atomic wire. According to classical physics, the absence of scattering centers implies that the electrical resistance must be exactly zero. However, when such a device is fabricated in a cleanroom, the measured resistance is found to be ~ 12.9 $k\Omega$

The origin of this resistance, in the absence of any defects, poses a conceptual paradox. This is precisely the paradox resolved by the Landauer formalism

I. The Mesoscopic Paradigm Shift and The Origin of R_0

Schematic block diagram of a standard two-terminal transport setup



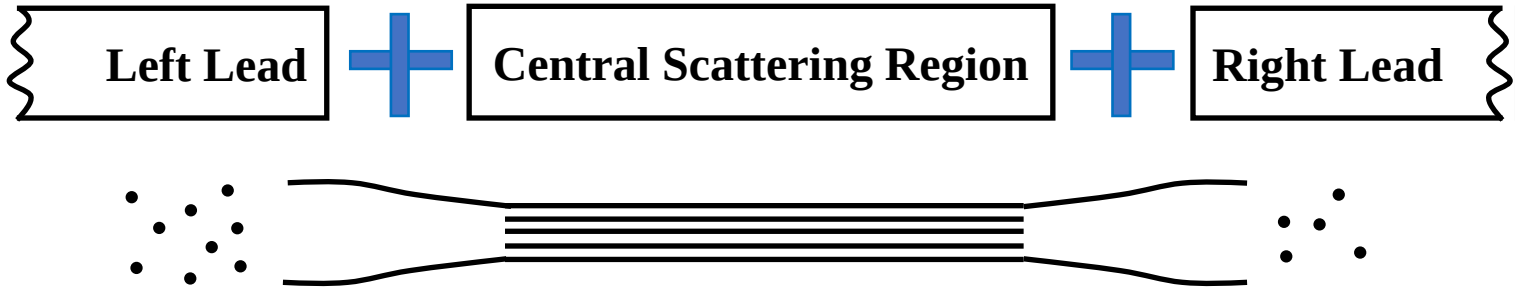
The Left/Right Reservoirs: Macroscopic, massive thermodynamic bodies at chemical potentials μ_L and μ_R. Electrons within them are fully thermalized according to the Fermi-Dirac distribution:

$$f_{L/R}(E) = \frac{1}{EXP\left(\frac{E-\mu_{L/R}}{k_B T}\right)+1}$$

To calculate transport rigorously, the open boundary system must be carefully isolated into three cleanly partitioned chunks. The actual nanostructure is placed in the 'Central Region'. Connected to it are 'Leads', which act as perfect quantum pipelines. These leads originate from massive macroscopic 'Reservoirs'. The reservoirs are critical: they possess an infinite capacity to supply or absorb electrons, maintaining a rigid thermal equilibrium within themselves even when a large electric current flows across the system

I. The Mesoscopic Paradigm Shift and The Origin of R_0

Origin of the Fundamental Quantum Resistance R_0



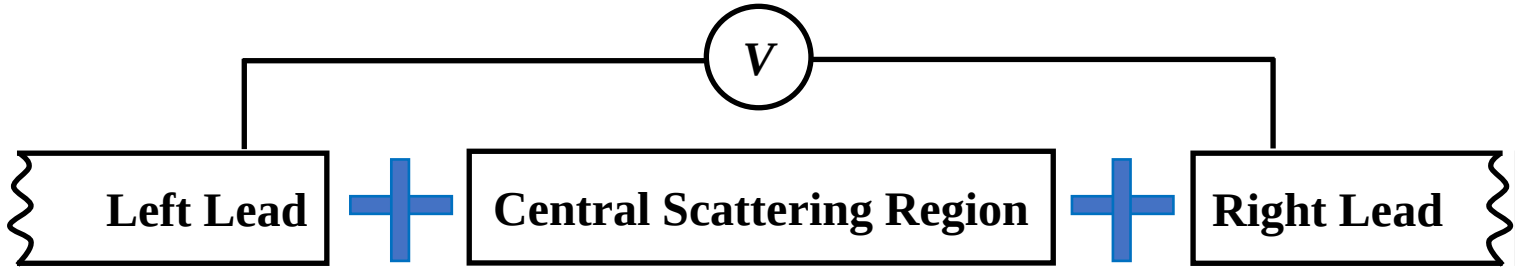
The sudden transition from an infinite number of electronic states in a macroscopic reservoir to a highly restricted, quantized number of modes inside the 1D nanoconductor. The resistance is not distributed along the length of the pristine channel; it is located entirely at the reflection planes where the leads meet the reservoirs

The 12.9 $k\Omega$ resistance mentioned before does not come from inside the wire; it originates from the injection interfaces. A massive highway tapering down into a tiny single-lane alleyway serves as an analogy. Electrons inside the reservoir have infinite degrees of freedom, but the 1D quantum channel can only accommodate a discrete number of transverse quantum states, called 'modes'. The reflection and backscattering generated at this geometrical bottleneck are what directly manifest as contact resistance

I. The Mesoscopic Paradigm Shift and The Origin of R_0

Injection Streams and State Counting

Applied external bias voltage relation: $\mu_L - \mu_R = eV_b$



Left Reservoir injections: Populate right-moving states ($k > 0$) inside the conductor up to energy

Right Reservoir injections: Populate left-moving states ($k < 0$) inside the conductor up to energy

Visual diagram of the energy spectrum $E(k)$ of a 1D conductor under bias, emphasizing the uncompensated window of current carriers between μ_L and μ_R

When a bias voltage V_b is applied across the contacts, the chemical potentials of the two reservoirs are shifted relative to each other such that $\mu_L - \mu_R = eV_b$. The left reservoir forces an influx of right-moving electrons into the wire, while the right reservoir pumps left-moving electrons. Below μR, the right-movers and left-movers completely balance out, yielding zero net current. The actual electrical current is carried exclusively by the uncompensated electrons residing inside the energy window $[\mu_L, \mu_R]$

II. Derivation of the Two-Terminal Landauer Formula

Derivation Step 1: Carrier Density and Velocity Interplay

The right-moving differential current (dI^+) in an energy window dE. The relation between spatial 1D carrier density dn^+ and the density of states per unit length $g_{1D}(E)$:

$$dn^+ = \frac{1}{2} g_{1D}(E) f_L(E) dE$$

1D spin-degenerate density of states formula: $g_{1D}(E) = \frac{2}{\pi}\left(\frac{dk}{dE}\right) = \frac{2}{\pi \hbar v(E)}$

Group velocity of the electron package: $v(E) = \hbar \frac{dE}{dk}$

The formal derivation of the Landauer formula begins with the definition of the macroscopic current contribution dI^+ within a tight energy window dE as the charge (e) multiplied by the spatial carrier density (dn^+) multiplied by the velocity of those carriers ($v(E)$). Examining the 1D density of states $g_{1D}(E)$, it is found to be inversely proportional to the group velocity. This implies that as electrons move faster, their density decreases proportionally

II. Derivation of the Two-Terminal Landauer Formula

Derivation Step 2: Cancelling the Velocity Term

Substitute the density of states $g_{1D}(E)$ directly into the raw current equation: $dI^+ = e \cdot dn^+ \cdot v(E)$

$$dI^+ = e \cdot \left[\frac{1}{2}\frac{2}{\pi\hbar v(E)} f_L(E)dE\right] \cdot v(E)$$

Show explicitly on the board how $v(E)$ cancels out cleanly: $dI^+ = \frac{2e}{h} f_L(E)dE$

The group velocity $v(E)$ in the numerator cancels out the group velocity term embedded within the 1D density of states in the denominator. This is a profound moment in quantum transport physics. It proves that for a single 1D channel, the amount of current injected per unit energy is an absolute universal constant ($\frac{2e}{h}$), entirely independent of the material properties, band structure, or chemical composition of the conductor

II. Derivation of the Two-Terminal Landauer Formula

Derivation Step 3: Integrating the Net Uncompensated Flux

The net current I is the difference between the total right-moving stream (I^+) and the left-moving stream (I^-):

$$I = I^+ - I^- = \int_{-\infty}^{\infty} \frac{2e}{h}[f_L(E) - f_R(E)]dE$$

At zero temperature (T = 0 K), the Fermi functions become sharp step functions. The integral collapses onto the bias window boundaries:

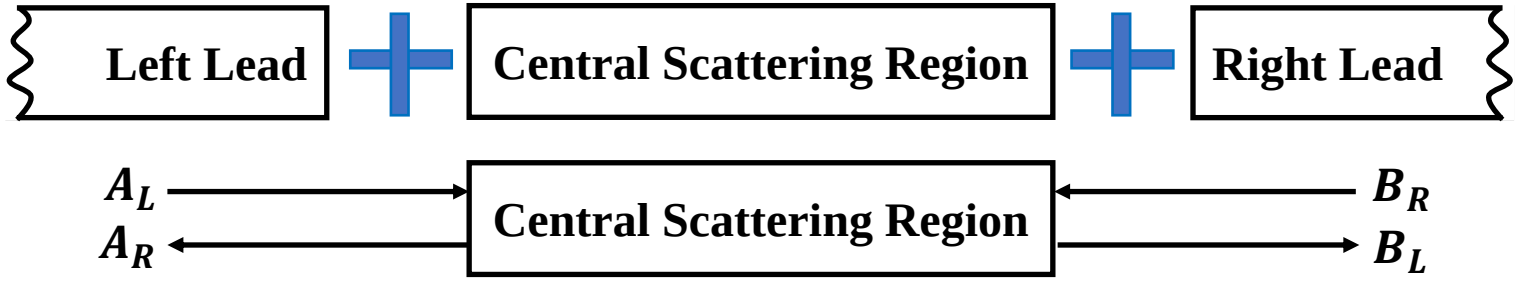
$$I = \int_{\mu_R}^{\mu_L} \frac{2e}{h} dE = \frac{2e}{h}(\mu_L - \mu_R) \xrightarrow{\mu_L - \mu_R = eV_b} I = \frac{2e^2}{h} V_b$$

By taking the difference between the two counter-propagating injection streams, the total net current passing through the perfect ballistic wire is obtained. Under absolute zero temperature, the integration is incredibly straightforward because the difference between the Fermi distributions is exactly 1 inside the window $[\mu_L, \mu_R]$ and 0 everywhere else. This yields the quantized conductance relationship: $I = G_0 V_b$

II. Derivation of the Two-Terminal Landauer Formula

Scattering Matrix (*S*)

Transitioning from a perfect wire to a realistic wire containing a potential barrier or scattering center—The Quantum Scattering Problem



Definition of incoming and outgoing plane wave amplitudes at the interfaces: A_L, B_L(left lead) and A_R, B_R(right lead)

The Scattering Matrix (S) structure: $\begin{pmatrix} B_L \\ B_R \end{pmatrix} = S \begin{pmatrix} A_L \\ A_R \end{pmatrix} = \begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} A_L \\ A_R \end{pmatrix}$

- **Transmission amplitude coefficients — t, t'**
- **Reflection amplitude coefficients — r, r'**

Up to this point, perfect transmission (100%) has been assumed for every injected electron. A real obstacle — such as a localized defect or impurity barrier — is now introduced in the central region. Mathematically, this is treated using an *S* matrix. The *S* matrix acts as a linear operator that maps all incoming wave amplitudes (*A*) into all backscattered or transmitted outgoing wave amplitudes (*B*)

II. Derivation of the Two-Terminal Landauer Formula

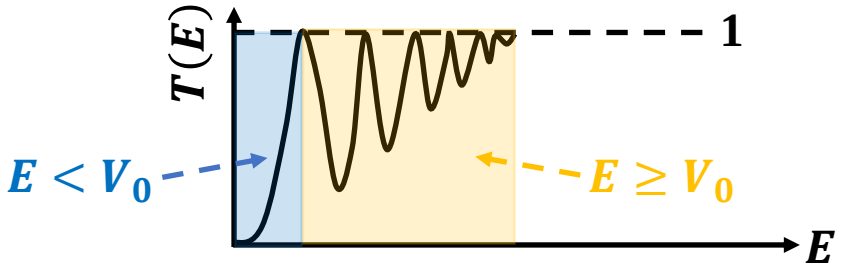
Transmission Probability $T(E)$

Definition of the dimensionless Transmission Probability $T(E)$: $T(E) = |t(E)|^2$

Definition of the Reflection Probability $R(E)$: $R(E) = |r(E)|^2$

Conservation of particles/Unitarity of the S matrix ($ss^\dagger = I$)

Visual plot examples of $T(E)$ *v.s.* E for a rectangular barrier (quantum tunneling below the barrier height, over-barrier resonances above it)



The absolute square of the complex transmission amplitude $t(E)$ yields the real, dimensionless probability $T(E)$. This value ranges strictly between 0 (complete block) and 1 (perfect transmission). Since the system is stable and contains no electron sinks or sources inside the scattering spot, particle conservation requires that any electron not transmitted be reflected. Thus, $T(E) + R(E) = 1$

II. Derivation of the Two-Terminal Landauer Formula

The Two-Terminal Landauer Formula

Integrating the transmission probability $T(E)$ directly into the phase-space current equation:

$\frac{2e}{h}$ — **Maximum quantum current capacity per unit energy**

$$I = \frac{2e}{h}\int_{-\infty}^{\infty} T(E)[f_L(E) - f_R(E)]dE$$

$T(E)$ — **Quantum mechanics fingerprint (wave mechanics, tunneling)**

$[f_L(E) - f_R(E)]$ — **Thermodynamics fingerprint (thermal broadening, voltage bias)**

This is the Landauer Formula for a single-channel conductor. It is an exceptionally beautiful equation because it cleanly isolates the two defining pillars of mesoscopic transport. The term $T(E)$ contains all the complex quantum mechanics of the device — such as tunneling states, phase interference, and geometry. Meanwhile, the difference in Fermi functions contains all the thermodynamics — capturing the effect of thermal smearing and the external voltage drive

III. Generalization to Multi-Terminal Networks and Topological Applications

Multi-Channel Generalization: Transverse Subbands

Moving from a purely 1D wire to a wider 2D nanoribbon or 3D nanowire

Quantum confinement across the transverse directions generates discrete subbands (modes)

Let $M(E)$ be the total number of active transverse modes available at energy E

Total transmission is now a summation over all open channels: $\mathcal{T}(E) = \sum_{n=1}^{M(E)} T_n(E)$

The generalized multi-channel Landauer formula: $I = \frac{2e}{h}\int_{-\infty}^{\infty} \mathcal{T}(E)[f_L(E) - f_R(E)]dE$

Real nanodevices are rarely strictly 1D; they have a finite width and thickness. This structural confinement splits the electronic spectrum into multiple parallel transport tracks, which we call 'transverse modes'. If multiple modes are active at a given energy, they act like parallel wires. The total transmission function $\mathcal{T}(E)$ is simply the sum of the individual transmission probabilities $T_n(E)$ across all open channels

III. Generalization to Multi-Terminal Networks and Topological Applications

The Linear Response Regime: $V_b \to 0$

Definition of the Linear Response limit: The applied bias voltage V_b is small compared to the energy scales over which $T(E)$ varies: $eV_b \ll k_B T$ or $eV_b \ll \Delta E_{resonance}$

Taylor expansion of the Fermi-Dirac distribution function difference around the equilibrium chemical potential E_F:

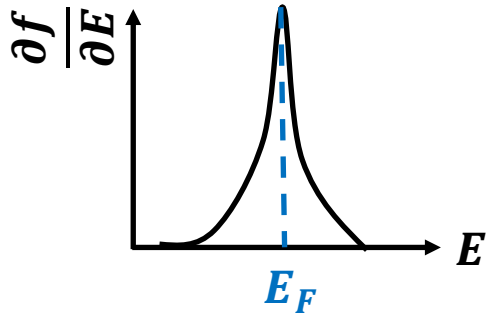
$$f_L(E) - f_R(E) \approx f\left(E - \frac{eV_b}{2}\right) - f\left(E + \frac{eV_b}{2}\right) \approx \left(-\frac{\partial f}{\partial E}\right) eV_b$$

Linear Conductance ($G = {}^{I}/_{V_b}$) equation: $G = \frac{2e^2}{h} \int_{-\infty}^{\infty} \mathcal{T}(E) \left(-\frac{\partial f}{\partial E}\right) dE$

In many electrical characterization experiments, a very tiny AC or DC voltage is applied to avoid driving the system out of its linear regime. Mathematically, as V_b approaches zero, the difference between the two shifted Fermi distributions is approximated by a derivative. The term $-\frac{\partial f}{\partial E}$ is known as the Fermi derivative window

III. Generalization to Multi-Terminal Networks and Topological Applications

The Physical Role of the Fermi Derivative Window: $-\frac{\partial f}{\partial E} = \frac{1}{k_B T} \frac{EXP\left(\Delta E / k_B T\right)}{\left[EXP\left(\Delta E / k_B T\right)+1\right]^2}$, **where** $\Delta E = E - E_F$



Full Width at Half Maximum (FWHM) of the curve is roughly 3.5 $k_B T$

Behavior at $T = 0\ K$**: The curve sharpens into a Dirac Delta function** $\delta(E - E_F)$

This derivative function $-\frac{\partial f}{\partial E}$ **behaves exactly like an energy sampling window or filter. It forms a clean, symmetric bell curve centered tightly at the Fermi energy** E_F**. At room temperature (300** K**), this window has a width of roughly 100** meV**. Any electronic features (like band gaps or resonant states) located outside this narrow window cannot contribute to linear electrical conduction because there are either no electrons available to move or no empty states to receive them**

III. Generalization to Multi-Terminal Networks and Topological Applications

The Zero-Temperature Limit: Pure Conductance Quantization:

Substitute $-\frac{\partial f}{\partial E} = \delta(E - E_F)$ **into the linear conductance formula:** $G = \frac{2e^2}{h}\int_{-\infty}^{\infty} \mathcal{T}(E)\delta(E - E_F)dE$

Evaluation of the delta integral: $G = \frac{2e^2}{h}\mathcal{T}(E_F) = G_0\mathcal{T}(E_F)$

For a perfectly ballistic multi-channel wire ($T_n = 1$ **for all open channels):** $G = \frac{2e^2}{h}M(E_F)$

Cooling the system down to absolute zero causes the sampling window to collapse into an infinitely sharp Dirac Delta function. The integral vanishes, and the conductance simplifies into a beautifully compact expression: the conductance is determined solely by the transmission value evaluated right at the Fermi level. If the system is perfectly ballistic, the conductance step-increases as an integer multiple of G_0, tracking the number of active modes

III. Generalization to Multi-Terminal Networks and Topological Applications

The Deep Philosophical Paradigm Shift of Landauer Theory

Classical viewpoint: Classical conduction is caused by internal local electric fields accelerating local carrier pools, balanced by local scattering events ($J = \sigma E$)

Landauer viewpoint: Conduction is fundamentally a quantum mechanical transmission/scattering problem across an open system boundary

'Conductance is Transmissivity': The calculation does not determine how much electrons accelerate inside the device; instead, it calculates their probability of successfully passing through it

A step back reveals the underlying philosophy. In classical physics, an applied electric field accelerates the local pool of electrons inside the material. The Landauer approach requires a complete change in worldview. What happens 'inside' the material in terms of local acceleration is no longer the focus. Instead, the entire device is treated as an optical obstacle and the electron as a wave. Conductance ceases to be an intrinsic bulk property; it becomes simply the overall probability that a wave injected at one end successfully emerges at the other

III. Generalization to Multi-Terminal Networks and Topological Applications

Multi-Terminal Circuit Topologies

Real-world configurations: Hall bars, multi-tip probes, quantum ring interferometers, *Y* junction splitters

Labeling protocol: Let index *p* represent the target probe/terminal, and index *q* represent the source probe/terminal.

V_p, V_q: Applied electrostatic voltages at individual terminals

$T_{qp}(E)$: Microscopic transmission probability for an electron to be injected from terminal *p* and successfully collected at terminal *q*

A major strength of the Landauer framework is its ability to scale beyond simple two-terminal devices. Many advanced quantum experiments — such as those measuring the Quantum Hall Effect — employ geometries with four, six, or more contact probes. In modeling such systems, the terms are generalized: T_{qp} represents the directional transmission probability of a quantum wave traveling from an injection probe *p* through the central junction into a collection probe *q*

III. Generalization to Multi-Terminal Networks and Topological Applications

Derivation of the Landauer-Büttiker Multi-Terminal Equations

Total current leaving terminal *p* consists of electrons injected into all other terminals *q* that end up inside terminal *p*, minus the electrons terminal *p* injects out into the world

Applying linear response simplifications to the multi-terminal particle flux yields the Landauer-Büttiker Formalism:

$$I_p = \frac{2e^2}{h} \sum_q \left[T_{qp} V_p - T_{pq} V_q \right]$$

Summation constraints and Current Conservation requirement: $\sum_p I_p = 0$

Markus Büttiker expanded Landauer's single-channel equation into this multi-terminal system of linear equations. The net current I_p entering or exiting a given probe *p* is a balance of trade: it is the sum of all current packages successfully transmitted from probe *p* out to all other terminals *q*, balanced against the inbound streams arriving at probe *p* from the rest of the network

III. Generalization to Multi-Terminal Networks and Topological Applications

The Unitarity and Sum Rules of Transport Matrices

Matrix Constraints: Reciprocity and Coefficients Sum Rules

Because electrons cannot vanish into thin air, the total sum of coefficients must satisfy strict conservation bounds:

Total modes leaving probe p: $\sum_{q \neq p} T_{qp} = M_p$

Alternative compact formulation of the Büttiker equation: $I_p = \frac{2e^2}{h}\left[M_p V_p - \sum_{q \neq p} T_{pq} V_q\right]$

Onsager Reciprocity Relations under an external magnetic field B: $T_{pq}(B) = T_{qp}(-B)$

Because these equations describe a conservative quantum system, the coefficients cannot be arbitrary values. The total number of channels leaving a probe must exactly equal the sum of all reflections and transmissions distributed across the other probes. Furthermore, under an external magnetic field, time-reversal symmetry imposes Onsager reciprocity: if you flip the direction of the magnetic field, the transmission probability from probe p to q maps exactly onto the reverse path from q to p

III. Generalization to Multi-Terminal Networks and Topological Applications

Example: Edge State Transport in the Quantum Hall Effect — Chiral Edge Channels in a 4 Probe Hall Bar

Schematic of a 4 terminal Hall bar under a strong perpendicular magnetic field.

The concept of Chiral Transport: Electrons can only move in one direction along the physical boundary of the sample due to Lorentz force skipping orbits.

Microscopic Transmission assignments for this chiral system: ♦ $T_{21} = M, T_{32} = M, T_{43} = M$ **and** $T_{14} = M$

♦ **All other cross-terms** $T_{pq} = 0$ **(Zero backscattering)**

Solve the matrix system to show that the longitudinal resistance $R_{xx} = 0$**, and the Hall resistance is perfectly quantized:**

$$R_{xy} = \frac{h}{e^2}\frac{1}{M}$$

In the Quantum Hall Effect, a powerful magnetic field localizes electrons in the bulk, while those at the edges form robust, one-way streets called 'chiral edge states'. Traveling on one-way paths, these edge states cannot turn around and backscatter off defects. Inserting these perfect asymmetric transmission values into the Büttiker equations instantly yields the deduction: the longitudinal resistance drops to absolute zero while the Hall resistance locks onto incredibly precise universal plateaus



That's all for this session

Thank you for your attention and great participation